

MONTANA, Switzerland

WMO: 067240

Lat: 46.2986N

Lon: 7.4608E

Elev: 1508

StdP: 84.47

Time Zone: 1.00 (EUC)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-12.1	-10.1	-19.5	0.8	-8.3	-17.0	1.0	-6.1	7.7	2.4	6.8	1.7	1.2	30	0.576

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	9.7	25.2	15.1	23.4	14.5	21.9	14.0	16.0	23.0	15.3	21.7	14.5	20.5	2.9	250

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
13.5	11.6	17.8	12.7	11.0	17.2	12.0	10.5	16.9	50.6	23.2	48.2	21.9	45.9	20.6	19.1

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 7.1	5.8	5.0	-14.5	27.7	2.7	1.7	-16.4	28.9	-18.0	29.9	-19.6	30.8	-21.6	32.1		
(5)			-15.3	17.6	2.6	0.7	-17.2	18.1	-18.7	18.5	-20.1	18.9	-22.0	19.4		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	6.6	-1.8	-1.4	2.0	5.6	9.6	13.6	15.3	15.0	11.3	7.9	2.4	-0.6		
	DBStd	7.29	3.94	4.48	4.10	3.96	3.64	3.91	3.44	3.55	3.49	3.80	4.48	4.18		
	HDD10.0	1808	364	318	248	141	51	12	3	4	26	83	229	329		
	HDD18.3	4298	623	552	506	383	270	149	103	113	212	322	478	587		
	CDD10.0	578	0	0	0	7	39	119	169	159	65	19	1	0		
	CDD18.3	26	0	0	0	0	6	11	10	10	0	0	0	0		
	CDH23.3	155	0	0	0	0	1	36	59	58	1	0	0	0		
	CDH26.7	13	0	0	0	0	0	4	4	6	0	0	0	0		
(14) Wind	WSAvg	1.8	1.7	1.8	2.0	2.3	2.0	1.8	1.8	1.6	1.5	1.7	1.9	1.8		
(15) Precipitation	PrecAvg	1046	75	75	67	71	99	106	118	112	76	69	84	95		
	PrecMax	1394	240	309	195	146	186	204	233	188	142	159	215	259		
	PrecMin	829	9	10	18	22	24	52	58	30	35	2	1	0		
	PrecStd	150	55	68	45	31	45	36	41	40	32	36	53	60		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	9.3	10.6	13.7	18.2	22.4	26.7	27.0	27.4	22.4	18.8	13.8	9.7		
		MCWB	2.7	3.2	5.8	8.7	12.5	15.1	15.8	15.4	13.9	10.8	6.5	2.7		
	2%	DB	6.4	8.1	11.7	15.6	19.6	24.1	25.0	25.0	20.5	16.6	11.3	7.6		
		MCWB	1.4	2.3	4.8	7.8	11.3	14.5	15.0	15.0	13.2	9.6	5.5	2.3		
	5%	DB	4.7	5.9	9.8	13.6	17.5	22.2	23.2	23.0	18.5	14.8	9.7	6.1		
		MCWB	1.0	1.3	4.0	6.6	10.2	13.9	14.6	14.5	12.3	9.0	5.0	1.8		
	10%	DB	3.5	4.2	7.8	11.6	15.6	20.3	21.6	21.2	16.7	13.2	8.3	4.8		
		MCWB	0.4	0.4	2.9	5.7	9.2	13.1	14.0	14.0	11.6	8.3	4.2	1.1		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	4.2	4.4	6.7	9.3	13.4	16.5	17.1	16.9	14.9	12.0	8.0	4.8		
		MCDB	6.3	8.4	12.4	16.7	20.6	23.9	24.0	24.5	20.3	16.4	11.4	7.0		
	2%	WB	2.8	2.9	5.3	8.1	11.9	15.4	16.1	15.9	13.8	10.8	6.6	3.5		
		MCDB	5.0	6.3	10.2	14.6	18.1	22.6	23.1	22.5	19.0	14.8	10.1	5.9		
	5%	WB	1.7	1.9	4.3	7.1	10.8	14.4	15.3	15.1	12.8	9.8	5.6	2.6		
		MCDB	4.0	5.2	8.7	12.5	16.2	20.8	21.7	21.3	17.5	13.4	8.7	5.0		
	10%	WB	0.7	0.8	3.4	6.1	9.8	13.5	14.5	14.4	11.9	9.0	4.6	1.7		
		MCDB	3.0	3.8	7.1	10.8	14.7	19.2	20.3	20.0	16.0	12.4	7.6	4.1		
(35) Mean Daily Temperature Range	5% DB	MDBR	5.6	6.6	7.7	8.2	8.7	9.4	9.7	9.1	8.3	7.3	5.8	5.6		
		MCDBR	6.8	8.4	9.7	10.6	11.1	11.6	11.6	11.5	10.1	9.0	7.3	7.1		
	5% WB	MCWBR	4.4	5.2	5.5	5.5	5.4	5.4	5.2	5.0	5.3	5.2	4.4	4.4		
		MCWBR	6.0	7.7	8.7	10.0	10.2	10.8	10.8	10.4	9.2	8.1	6.8	6.0		
(40) Clear-Sky Solar Irradiance	taub	taub	0.237	0.261	0.304	0.343	0.361	0.365	0.363	0.351	0.328	0.305	0.268	0.237		
		taud	2.341	2.285	2.237	2.245	2.283	2.316	2.316	2.361	2.417	2.463	2.453	2.379		
	Ebn at Noon	Ebn at Noon	884	922	922	911	904	899	896	894	888	862	838	845		
		Edh at Noon	72	95	117	128	128	125	123	113	98	80	65	62		
(44) All-Sky Solar Radiation	RadAvg	RadAvg	1.43	2.31	3.42	4.38	5.00	5.94	5.90	5.14	4.03	2.62	1.49	1.18		
	RadStd	RadStd	0.16	0.32	0.47	0.60	0.43	0.54	0.49	0.44	0.34	0.28	0.21	0.18		

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only		N/A	N/A	N/A	+0.90	+0.41	+0.27	N/A	N/A	+60	+10
(47) Regional (109 neighbors)		N/A	N/A	N/A	+0.71	+0.40	+0.30	N/A	N/A	+57	+16

Nomenclature: See separate page